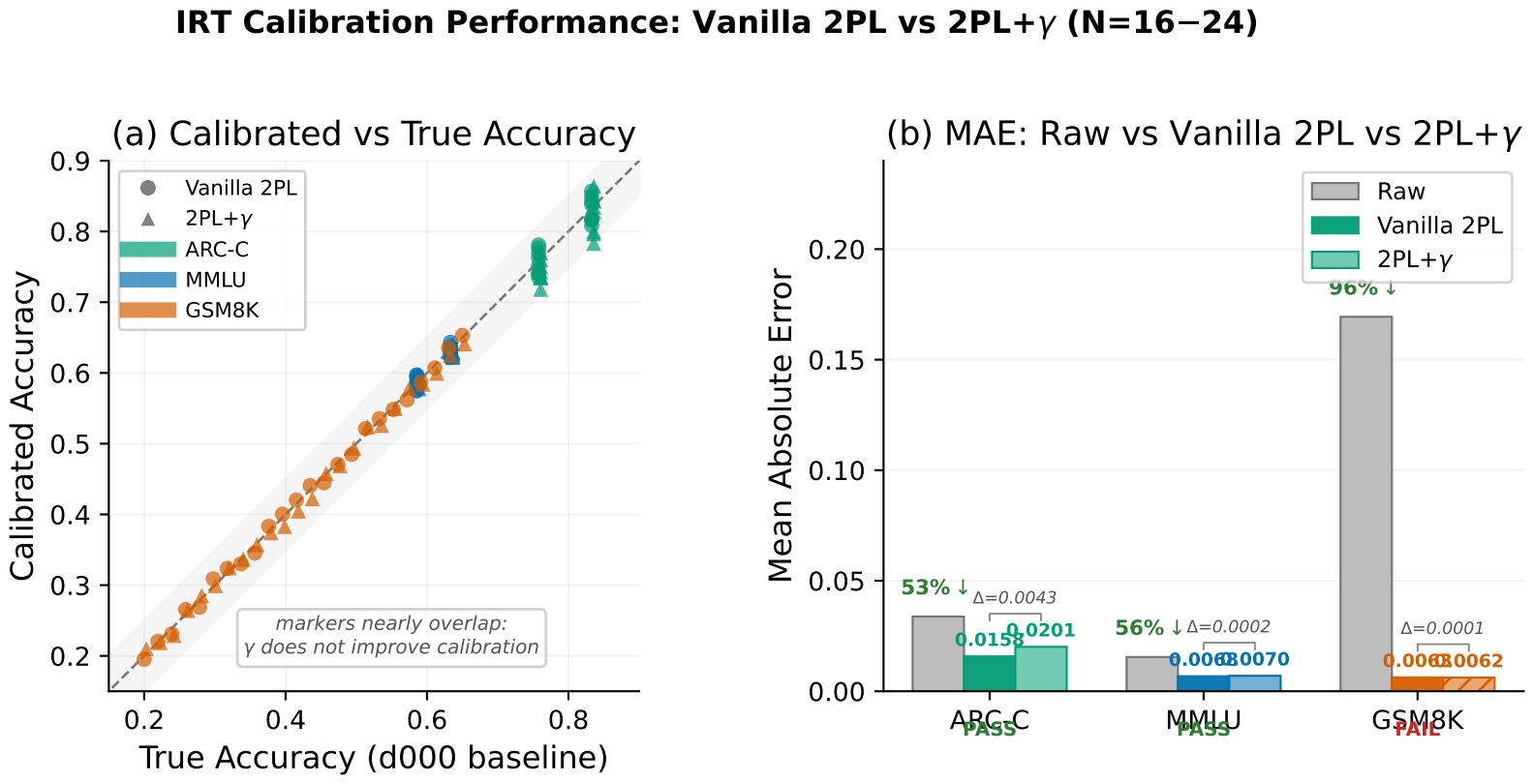
IRT Calibration Performance: Vanilla 2PL vs 2PL+γ (N=16−24)
(a) Calibrated vs True Accuracy
Vanilla 2PL
2PL+γ
ARC-C
MMLU
GSM8K
markers nearly overlap:
γ does not improve calibration
Calibrated Accuracy
True Accuracy (d000 baseline)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
(b) MAE: Raw vs Vanilla 2PL vs 2PL+γ
Raw
Vanilla 2PL
2PL+γ
Mean Absolute Error
0.00
0.05
0.10
0.15
0.20
53% ↓
Δ=0.0043
0.0158
0.0201
56% ↓
Δ=0.0002
0.0068
0.0070
96% ↓
Δ=0.0001
0.0062
0.0062
ARC-C
PASS
MMLU
PASS
GSM8K
FAIL